Reg. No. | | | | | | | | | | | [18MTFT201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE–2019

FOOD PROCESS ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
ii. Draw neat sketches wherever necessary.
iii. Make suitable assumptions wherever necessary.

1.	**a.**	Discuss rheological properties of liquid, solid and semi-solid food with appropriate mathematical formulations.	**10**
	b.	Discuss in brief various properties of foods, with an emphasis on the importance of physical properties during food processing and for equipment design.	**10**
2.	**a.**	What is screen effectiveness? Calculate the screen effectiveness when mass fraction of the feed, overflow and under-flow are 0.16, 0.711 and 0.021, respectively.	**10**
	b.	Comprehend on three different milling equipments used in the food industry with a neat sketch.	**10**
3.	**a.**	Outline the principles behind pascalization process and discuss on the construction of batch and semi continuous pascalization system.	**10**
	b.	Illustrate the basic principles involve in pulsed electric field (PEF) processing along with factors affecting PEF process and its effects on food.	**10**
		OR	
4.	**a.**	Summarize the working principles and applications of irradiation process in food industry.	**10**
	b.	Inspect the principle of UV and X-ray processing and list out its merits and demerits.	**10**
5.	**a.**	Summarize the applications of extruder and the process with neat diagram.	**10**
	b.	Examine the purpose of smoke application. Classify various smoking operations and discuss its merits and demerits.	**10**
		OR	
6.	**a.**	100 kg of carrots pieces were dehydrated in cabinet dryer. The initial moisture content was 84% (wb) and pieces are targeted to be dried up to moisture content 9% (wb). The critical moisture content is 29% (wb). The constant drying rate is 8 min. Estimate the total drying time for the products.	**10**
	b.	Comprehend on any four drying equipments, with a neat schematic diagram and merits.	**10**

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7. a. Formulate the principle behind infrared heating and ohmic heating and list out its applications in food processing. **10**

b A spherical food product is being frozen in an air-blast freezer. The initial product temperature is 45°C and the cold air 12°C. The product has a 5 cm diameter with density of 1000 kg/m^3, the initial freezing temperature is 1.25°C, the thermal conductivity of the frozen product is 1.2 W/(mK), and the latent heat of fusion is 250 kJ/kg. Convective heat-transfer coefficient is 50 W/(m^2 K). Compute the freezing time. **10**

OR

8. a. Highlight the working principles of microwave processing and explain the construction of a dielectric heater with neat sketch, application in food industry. **10**

b. Outline the principle behind the freeze drying operation with neat diagram. Dissect the merits of freeze drying over other drying methods. **10**